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PI Name:	Vitali, Rachel Ph.D.		
Project Title:	Scientific Physical and Operations Characterization (SPOC)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	48109-2117	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019 TRISH RFA-1901-PD Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Start/End dates changed to 9/1/2019-8/31/2020 per TRISH; original start date was 8/1/2019 and original end date was 7/31/2021 (Ed., 7/30/20) NOTE: End date changed to 7/31/2020 per TRISH; original end date was 7/31/2021 (Ed., 7/7/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Stirling, Leia Ph.D. (Mentor: Massachusetts Institute of Technology)		
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Task Description:	POSTDOCTORAL FELLOWSHIP The near future of human space exploration includes sending astronauts back to our moon and then on to Mars within the next few decades. These scientific investigations and pursuits will necessarily involve fieldwork to collect geological samples and survey new environments. However, there are many challenges to developing field operations for crew members during extravehicular activities. The processes that scientists utilize to achieve their fieldwork objectives in present-day terrestrial settings are not well-documented and measures do not currently exist to quantify operational performance for relevant tasks. Further, spacesuits will likely never be able to fully mimic a person's natural physical movements and given what little is known about the physical movements exhibited during scientific fieldwork conducted on Earth, this presents potentially significant limitations for future missions. The goal of this proposed research is to characterize physical movements during scientific fieldwork in terms of both biomechanical and motor control capabilities, which will be accomplished with noninvasive wearable sensor technology. Two studies are proposed, with the first study conducted in an unstructured environment with scientists conducting fieldwork. The second study is performed in a structured lab environment to evaluate techniques observed and measures defined from the first study. In the first study, outcomes will include using wearable technology to classify active versus inactive periods of time, to differentiate postures (e.g., upright, crouched, kneeling), and to assess the postures used during different tasks (e.g., traversing, documenting, sampling). In the second study, in addition to validating these classifiers, a study will be performed to assess the ability to perform relevant tasks and disambiguate these tasks using wearable sensors while using spacesuit gloves. Collectively, these two studies will have implications for the design and development of spacesuits, instruments, human-suit-tool interfaces, training protocols, and mission planning.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The impacts of this research are largely associated with a newly developed ability to study human behavior and performance in more naturalistic settings afforded by the unobtrusive wearable sensor technology employed in this work. By leveraging the portability and noninvasive nature of inertial measurement units (IMUs), expert scientists can more naturally conduct their unstructured fieldwork activities. Ultimately, this study design choice yields data that more accurately portrays important aspects of scientific fieldwork that are of interest to various stakeholders. However, raw IMU data (i.e., linear acceleration and angular velocity) require extensive metric and algorithm development for direct interpretations to make inferences about the scientific fieldwork. First, IMUs are suitable for activity classification given the unique kinematic signatures that are associated with various tasks. For example, gait is typically easily identifiable using IMUs attached to the feet given the cyclical nature of the movement. Automated activity classification significantly reduces the effort required to process large data sets, which inherently means more data can be collected and analyzed to produce a more representative depiction of scientific fieldwork. Furthermore, the results of the activity classification are useful in and of themselves. For example, knowing how much time is spent walking during a scientific fieldwork outing can help inform concept of operations and extravehicular activity (EVA) mission development, which take into account energy expenditure estimations for the astronauts completing those missions. Next, IMUs strategically attached to various body segments offer opportunities to develop operationally relevant metrics of performance as well as gather information about usage of specific joints. For example, IMUs attached to the sternum and upper arms facilitate extracting shoulder kinematics including joint range of motion that can be used to evaluate the risk of injury given the limitations of the current space suit. Joint kinematics can be analyzed at varying levels of specificity, meaning for an entire day trip down to for a specific task. These metrics of performance developed for expert scientists could also be extended to evaluating astronaut performance during training sessions. Further, IMUs allow for direct measurement of human kinematics within a spacesuit, which is currently impossible with more traditional video-based analysis. Finally, one of the critical aspects associated with conducting scientific fieldwork is the tools required to achieve specific tasks. However, the ways in which scientists employ these tools in a naturalistic setting will inherently be different from how astronauts will employ them in the future. It is important to catalogue how these tools are used, including the joint kinematics and kinetics required to use those tools, to maximize the probability of success for future EVA missions. The methods developed in this project offer opportunities to track how tools are used (e.g., intended and unintended modalities) and the kinematics associated with those modes of usage. For example, characterizing the cycles necessary to achieve a task given a specific tool design can be used to assess the tool's life expectancy (e.g., cycles until brittle fracture) for the range of thermal conditions expected during Artemis missions (e.g., permanently shadowed regions).
Task Progress:	The near future of human space exploration includes sending astronauts back to our moon and then on to Mars within the next few decades. The scientific investigations and pursuits integral to those missions will necessarily involve planetary fieldwork to collect geological samples or survey new environments, to name a couple examples. However, there are many challenges to developing field operations for crewmembers during extravehicular activities. For instance, the processes that scientists utilize to achieve their fieldwork objectives in present-day terrestrial settings are not well documented and measures do not currently exist to quantify operational performance for relevant tasks. Further, spacesuits will likely never be able to fully mimic a person's natural physical movements. Given what little is known about the physical movements exhibited during scientific fieldwork conducted on Earth, the suit-imposed mobility constraints present potentially significant limitations for future missions. The goal of this proposed research is to characterize physical movements during scientific fieldwork in terms of both biomechanical and motor control capabilities, which will be accomplished with noninvasive wearable sensor technology. Two studies are proposed, with the first study conducted in an unstructured environment with scientists conducting fieldwork. The second study will be performed in a structured lab environment to evaluate techniques observed and measures defined from the first study. Outcomes in the first study will include using wearable technology to classify active versus inactive periods of time, to differentiate postures, and to assess the postures used during different activities. For the second study, in addition to validating these classifiers, a study will be performed to assess the ability to perform relevant tasks and disambiguate these tasks using wearable sensors while using spacesuit gloves. To date, the key findings from this project are primarily associated with the first study focused on investigating how scientific fieldwork is physically conducted using only the unobtrusive wearable sensor technologies employed in this project, namely inertial measurement units (IMUs). A day trip to conduct scientific fieldwork can last for as many as 8-12 hours, producing large data sets that are unwieldy for manual processing and would otherwise limit these efforts. Thus, the activity classification for study 1 was automated in a two-stage process. The first stage focuses on identifying

	different motions (i.e., Global Walking, Local Walking, Partially Stationary, Completely Stationary), poses (i.e., upright, bent over, kneeling, crouching), and tool interactions (i.e., backpack, rock hammer, tablet) using only the data collected by the IMUs. The second stage infers what activities (i.e., reconnaissance, documentation, sampling) the person is engaged in based on the results from the first stage. An initial 90-minute data collection was performed at the NASA Johnson Space Center rockyard to evaluate the automated algorithms. The participant was outfitted with a full-body IMU array (i.e., head, sternum, sacrum, upper arms, forearms, hands, thighs, shanks, feet) and conducted scientifically relevant tasks throughout the site. Additional IMUs were attached to the backpack, rock hammer, tablet, and an Apollo-era scoop with extension handle. The results from this data collection demonstrate the feasibility of using wearable sensor technologies like IMUs to study and better understand that scientific fieldwork by first automatically parsing the data into operationally relevant segments. One significant implication of these findings relates to this newly developed ability to study various types of scientists (e.g., geologists, volcanologists, or geophysicists) in the real world to better understand the nature of their respective fieldwork. Next steps include validation of the activity classification methodology using video captured during the data collection, which is already underway to provide ground truth data for comparison. Furthermore, biomechanical metrics of performance like joint ranges of motion will be developed to characterize the capabilities required to conduct scientific fieldwork. Future data collections in collaboration with the U.S. Geological Survey are planned to outfit scientists going out on day trips to conduct their fieldwork in more authentic, unstructured conditions. Collectively, these two studies will have implications for the design and development of spacesuits, instruments, human-suit-tool interfaces, astronaut training protocols, and mission planning.
Bibliography Type:	Description: (Last Updated: 12/28/2020)
Papers from Meeting Proceedings	Vitali RV, Miller MJ, Stirling LA. "Methodology for the Scientific Physical and Operations Characterization (SPOC) of Terrestrial Fieldwork." 50th International Conference on Environmental Systems, Lisbon, Portugal, July 12-16, 2020. 50th International Conference on Environmental Systems, Lisbon, Portugal, July 12-16, 2020. ICES paper ICES-2020-163. , Jul-2020